

# Machine Learning for Trading in the Age of AI Agents

From market idea to a validated feature-and-label dataset

Stefan Jansen · *Machine Learning for Trading*, 3rd Edition

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# Build with me, then inspect the result

1. Keep the workshop repository and this deck open side by side.
2. Run each named notebook section when the green checkpoint slide appears.
3. Stop at the expected output and compare it with the slide before continuing.
4. Ask when your output differs. Do not debug silently while the session moves on.

The notebooks are prepared. The work today is understanding and checking each decision, not typing the entire pipeline from an empty file.

# One session, one break, one strategy

| Segment                                      | Mode                                         | Time          |
|----------------------------------------------|----------------------------------------------|---------------|
| Workflow, factors, and stack                 | Slides + discussion                          | 35 min        |
| Inspect the ETF data                         | Guided <code>01_data.ipynb</code>            | 15 min        |
| Build with a coding agent                    | Presenter-led demo                           | 20 min        |
| Engineer features and labels                 | Guided <code>02_features_labels.ipynb</code> | 30 min        |
| <b>Break</b>                                 | One short break                              | <b>10 min</b> |
| Model, backtest, research agents, next steps | Part 2                                       | 100 min       |

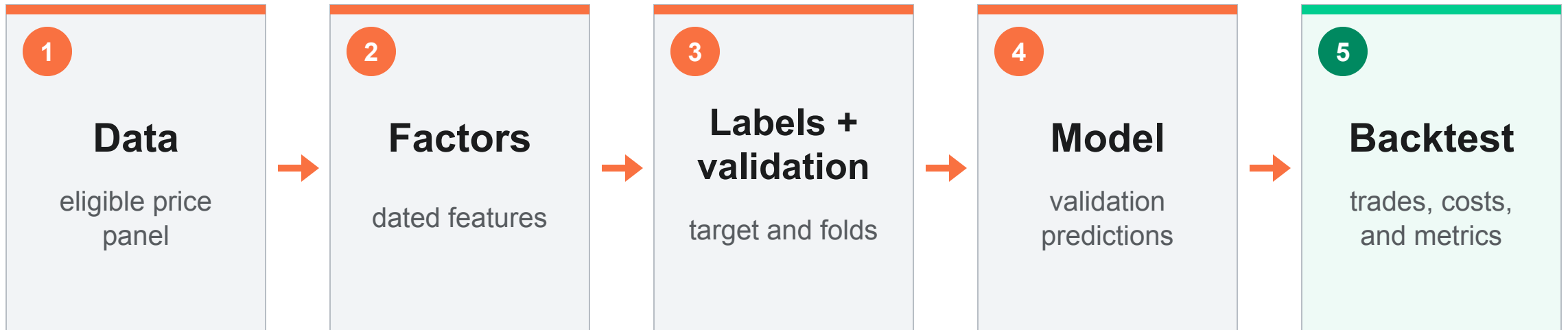


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# What you will leave with

- A complete LightGBM trading strategy built on 100 ETFs and daily data from 2006 to 2025.
- A point-in-time feature-and-label dataset with a manually verified 21-day forward return.
- Walk-forward predictions evaluated with both naive and HAC-corrected inference.
- A paired zero-cost and cost-aware backtest using identical signals and rebalance dates.
- Reusable notebooks, data, slide decks, and briefs for both agent demonstrations.

# Five decisions produce one backtest



# Agents assist; you approve the research

| Stage                 | An agent can assist                                      | You remain responsible for                |
|-----------------------|----------------------------------------------------------|-------------------------------------------|
| Alpha factors         | Propose hypotheses, summarize evidence, compare variants | Choosing a plausible mechanism            |
| Labels and validation | Flag leakage risk and conflicting horizons               | Approving the target and time split       |
| Model                 | Build a pipeline from a falsifiable specification        | Verifying inputs, outputs, and evaluation |
| Backtest              | Compare cost and turnover scenarios                      | Certifying the result                     |
| Live handoff          | Draft controls and monitoring checks                     | Deciding whether to deploy                |



# A testable factor begins as a specification

| Field              | Example: 252-day momentum                                    |
|--------------------|--------------------------------------------------------------|
| Mechanism          | Persistent sector flows may continue before reversing        |
| Formula            | 252-trading-day percent price change within each ETF         |
| Availability       | Uses closes available through the feature date               |
| Expected direction | Higher momentum predicts a higher 21-day forward return      |
| Failure state      | Sharp turning points can reverse before the feature responds |

# Research agents expand the candidate set

| A research agent can                                                                               | The decision still requires you to judge                     |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Propose factor hypotheses from a market observation                                                | Whether a mechanism exists beyond sample correlation         |
| Summarize evidence for a candidate                                                                 | Whether that evidence transfers to this universe and horizon |
| Compare variants on a common panel                                                                 | Whether the comparison itself is fair                        |
| The useful output is a smaller set of testable specifications, not a longer list of trading ideas. |                                                              |





# Eight features cover four market properties

| Family         | Features used today                  | Property represented                    |
|----------------|--------------------------------------|-----------------------------------------|
| Momentum       | 21, 63, 126, and 252-day returns     | Persistence across one to twelve months |
| Volatility     | 21 and 63-day realized volatility    | Current risk regime                     |
| Mean reversion | RSI over 14 days                     | Position within the ETF's recent range  |
| Liquidity      | 21-day dollar-volume percentile rank | Relative tradability on the same date   |

# Verify the dataset before using it

1. Run through **Sanity checks before we build anything**.
2. Confirm the symbol count and observed date range.
3. Continue through **Point-in-time eligibility**.
4. Stop after the eligible-symbol count by year.

Expected: 100 symbols, 2006-01-03 through 2025-12-31; XLC is the only ETF with less than ten years of observations.

# The file contains 100 ETFs across 20 years

100

ETF symbols

2006

first workshop date

2025

last workshop date

- The data ships with the repository, so the workshop does not depend on a live data service.
- Coverage differs by symbol: XLC begins in 2018, while many ETFs cover the full workshop window.
- File presence alone does not make an ETF eligible for every historical decision.

Source: executed output, student-repo/notebooks/01\_data.ipynb

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# Point-in-time eligibility changes the universe

- The 100-symbol file was selected with a backward-looking liquidity rule.
- The eligibility table records whether each symbol cleared that rule in each calendar year.
- Applying it before feature construction retains **84.2%** of all observed (symbol, day) rows.
- A 2006 model must not trade an ETF that became investable years later.

Source: executed output, student-repo/notebooks/02\_features\_labels.ipynb

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# Three libraries carry the trading pipeline

| Library         | Responsibility in this workshop                              |
|-----------------|--------------------------------------------------------------|
| ml4t-engineer   | RSI and point-in-time feature construction                   |
| ml4t-diagnostic | Walk-forward cross-validation and HAC-corrected IC inference |
| ml4t-backtest   | Event-driven execution with commissions and spread           |

pandas handles the panel, LightGBM estimates the return model, and Polars supplies the backtest engine's columnar inputs. Everything installs from the attendee repository.

PRESENTER-LED DEMONSTRATION · 20 MIN

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# A coding agent builds against a falsifiable brief

Watch the first attempt, inspect two failure-prone steps, and compare the result with the prepared notebooks.

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# Precision determines what the agent can verify

The live brief requires the agent to:

1. Apply point-in-time eligibility before any feature is built.
2. Build the named eight features using information available by the feature date.
3. Stop before executing the 21-day forward-return label.
4. Use walk-forward validation with `label_horizon=21`.
5. Stop before reporting naive and HAC-corrected IC statistics.

The complete brief stays in `docs/coding_agent_brief.md` for reuse after the workshop.

# Two checkpoints expose plausible wrong answers

| Stop                   | Inspect                                             | Failure it can reveal                         |
|------------------------|-----------------------------------------------------|-----------------------------------------------|
| Before labels run      | Return direction, shift direction, and manual check | A target attached to the wrong date           |
| Before evaluation runs | Splitter, purge horizon, IC series, and HAC horizon | Future information or overstated significance |

An error-free run does not establish that either step is correct.



# Apply eligibility and build eight features

1. Run through **Point-in-time eligibility, applied**.
2. Confirm that 84.2% of observed rows remain eligible.
3. Run **Features** and inspect the eight-column summary.
4. Identify which features operate within a symbol and which operates across symbols.

Expected: four momentum features, two volatility features, RSI(14), and cross-sectional dollar-volume rank.

# Feature construction uses two axes

## Within each ETF

- Momentum over four horizons
- Realized volatility over two horizons
- RSI with Wilder smoothing

## Across ETFs on each date

- 21-day average dollar volume
- Percentile rank within the eligible cross-section

Mixing the two axes changes the meaning of the liquidity feature.

# One line attaches the future outcome

```
g["fwd_ret_21d"] = g["close"].pct_change(21).shift(-21)
```

- `pct_change(21)` compares the current close with the close 21 rows earlier.
- `shift(-21)` attaches that realized return to the feature row 21 trading days earlier.
- The sign and shift direction require a price-level check; code review alone is insufficient.

# Prove the label against observed prices

1. Run **Labels: forward returns**.
2. Run **Prove the label doesn't leak**.
3. Compare the manual and computed SPY return.
4. Stop when the assertion passes.

Expected: manual 0.002639 , computed 0.002639 , followed by the no-off-by-one confirmation.

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# Random splitting violates the trading clock

- Consecutive 21-day labels share roughly 20 days of their return paths.
- A random split scatters overlapping observations across training and validation sets.
- Walk-forward validation keeps training before testing.
- Purging removes training labels whose outcome windows cross the validation boundary.

The label horizon must agree with both the purge rule and the later HAC correction.

## Save the model-ready panel

1. Run **Assemble and save the model dataset**.
2. Confirm the final row count, symbol count, and date range.
3. Verify that `data/model_dataset.parquet` now exists.

Expected: 369,159 rows, 99 symbols, 2008-01-03 through 2025-12-01.

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## Part 1 produced a falsifiable dataset

1. The universe is filtered at the decision date before features are built.
2. Each feature identifies its computation axis and available information.
3. The forward-return label matches a manual price calculation.
4. The saved panel is ready for leakage-aware walk-forward evaluation.

10-MINUTE BREAK

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# Save your notebook state

We resume with `notebooks/03_model_training.ipynb` and build out-of-sample predictions.